

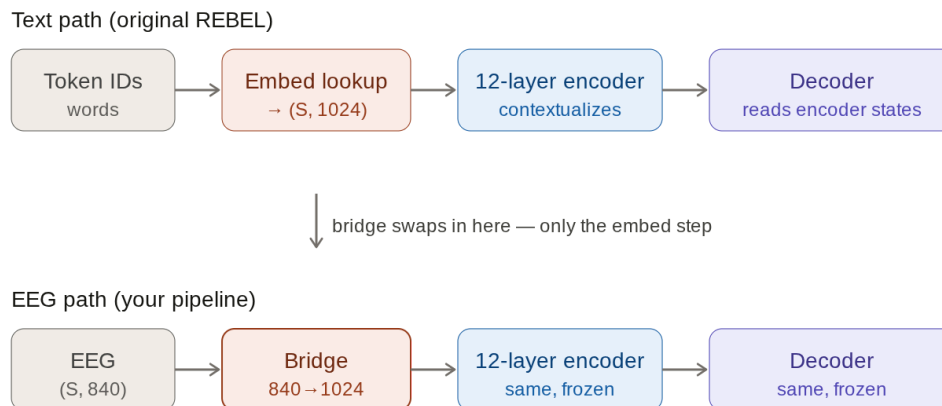
Final Experiments:

Research Question 2 (RQ2):

We want to understand how good the Bridge layer should be, meaning, how close it needs to get to the true sentence vector embedding that the REBEL knows to extract a triplet from.

In other words, how demanding is the REBELs Decoder?

Structure Visualization:



S = number of words in a sentence

E1 - Text-teacher upper bound + noise-interpolation curve (RQ2):

We are tackling this question by corrupting the Encoder output - interpolate the true embedding after attention with noise:

$$\text{states}(\alpha) = \alpha \cdot \text{text} + (1-\alpha) \cdot \text{noise}$$

And the state will pass into the REBELs Encoder to generate triplets

What are we measuring?

We are measuring 4 metrics - 2 measures the triplets and 2 measures the alignment

1. **F1** that uses **extract** - its the exact same tool used to generate the gold triplets - and so its best result (when $\alpha=1$) is 1
2. **F1** that uses **delinearize** - like on train eval
3. **pooled cos** - take the corrupted states ($S \times 1024$), average them down to one 1024-vector, do the same to the gold states, and measure cosine similarity between those two single vectors. It's "how aligned is the overall direction, after collapsing all S positions into one summary."
1 = identical direction, 0 = orthogonal.
4. **retrieval@1 (InfoNCE)** Setup: you have 67 sentences. For each one, pool its corrupted states into a single vector, pool every sentence's gold states into single vectors, and ask — does this sentence's corrupted vector retrieve its own gold vector as the nearest match, out of all 67?
retrieval@1 = the fraction of sentences where the correct gold vector ranks #1. 1.0 means every corrupted sentence still points closest to its own gold partner.
State = Encoder output

Background Explanation - Contrastive Learning:

Instead of training against a true gold label, the training involves pushing and pulling.

Everything is a multi-dimensional vector, and you train the model so that semantically close things will have a close embedding representation, and vice versa.

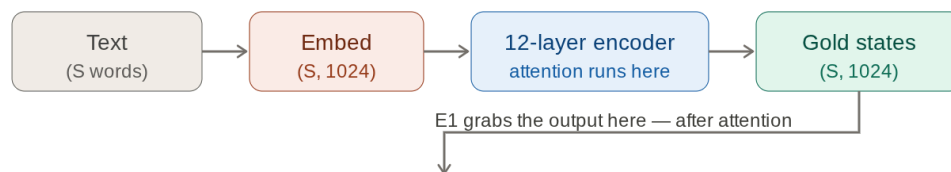
Why are we using it?

We are using teacher forcing - meaning when training, the decoder gets the prev gold embeddings (states). So it won't need much from the bridge in order to lower the CE loss.

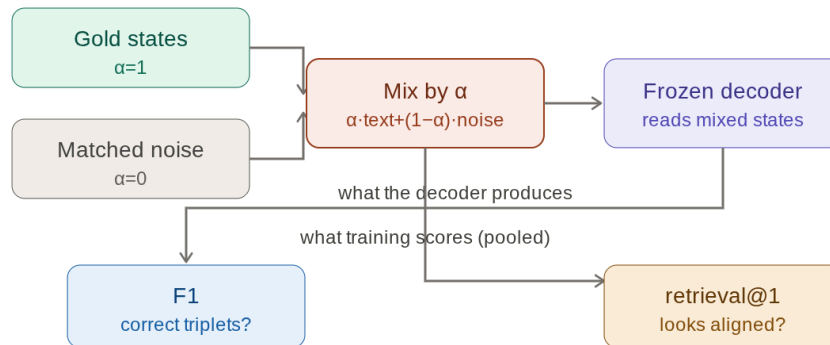
Using InfoNCE lets us add to the loss function a part that depends only on the state and wants to pull close to the gold one and push apart from the others.

Experiment Visualization:

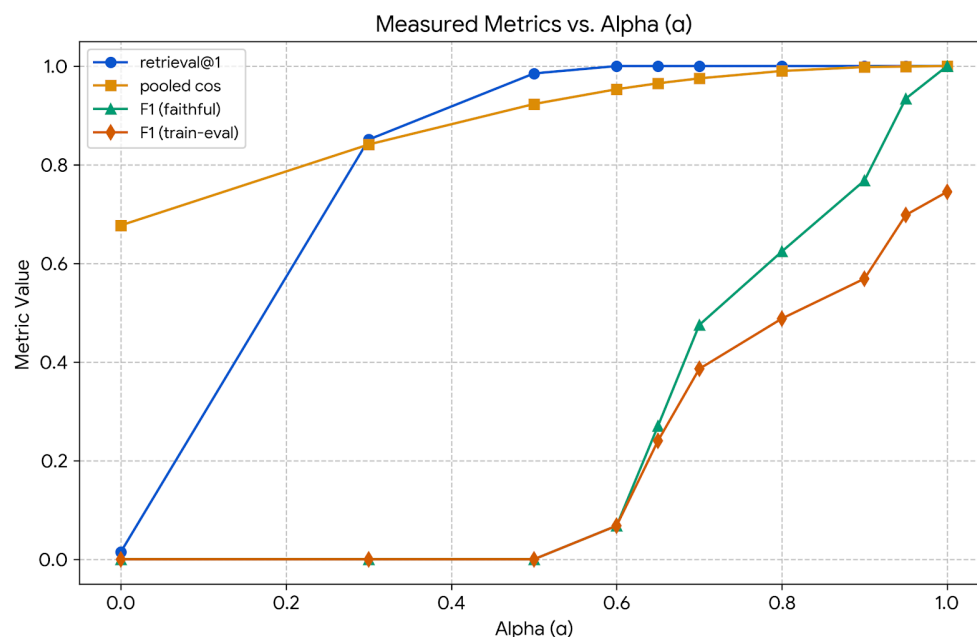
Stage 1 — produce gold encoder output (attention happens here)



Stage 2 — corrupt the states and decode



Experiments Results:



As we can see - F1 metric is very hared - drops to 0 quite fast, while the other 2 metrics are flat on 1 for a lot of noise margin.

So the problem is our loss function is flat for a big changes on the EEG (simulated as noise) - so no matter how many epochs we will have, there will be no improvement.

Bottom line - the alignment is a perfect 1 when F1 is 0, so it doesnt help with learning - **will be tackled on E7.**

RQ1 - Is there a useful signal in the EEG Processed Data?

E4 - Signal Probes (REBEL free):

Main Idea:

Stripping away the entire model (no Bridge, no REBEL), added a simple linear classifiers directory to the raw EEG features.

The mission - given an EEG data, can a linear classifier decide which of the 67 test sentences the person was reading? (called "sentence retrieval")

Metric:

The linear classifier takes one EEG recording, and given that, provides a ranked list: meaning which sentence he most likely thinks it is, which is the second, and so on.

A success (hit) is when the true gold sentence is somewhere in the top 10 sentences in the ranked list.

Top 10 hit rate is the fraction of EEG recordings that got "hit".

This is our metric because we want to see if the EEG recording even points us in the right direction.

Metric baseline:

$10/67 \approx 0.149$ is the baseline. This is the chance for random guessing, and this is what we need to beat.

Another metrics:

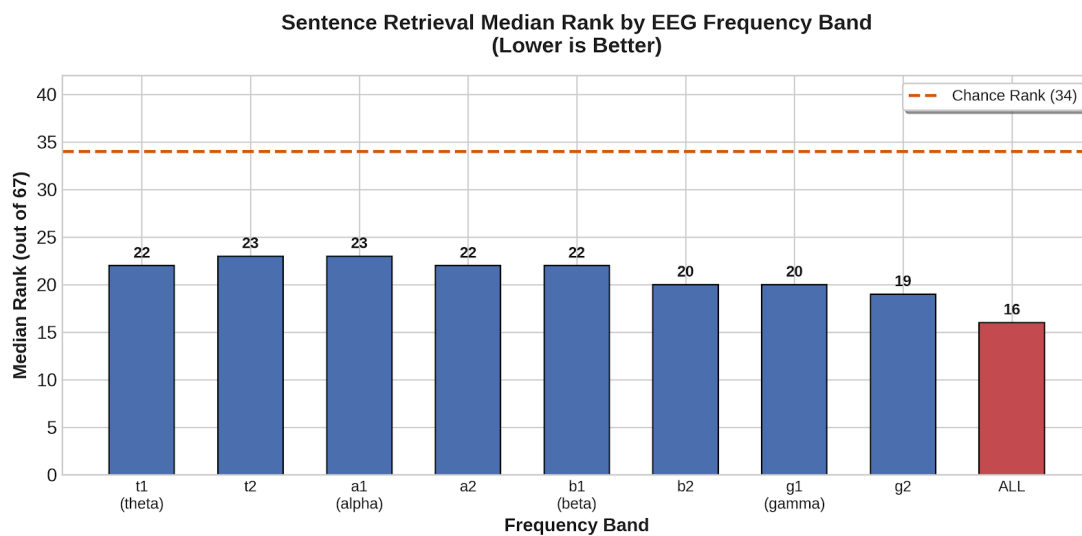
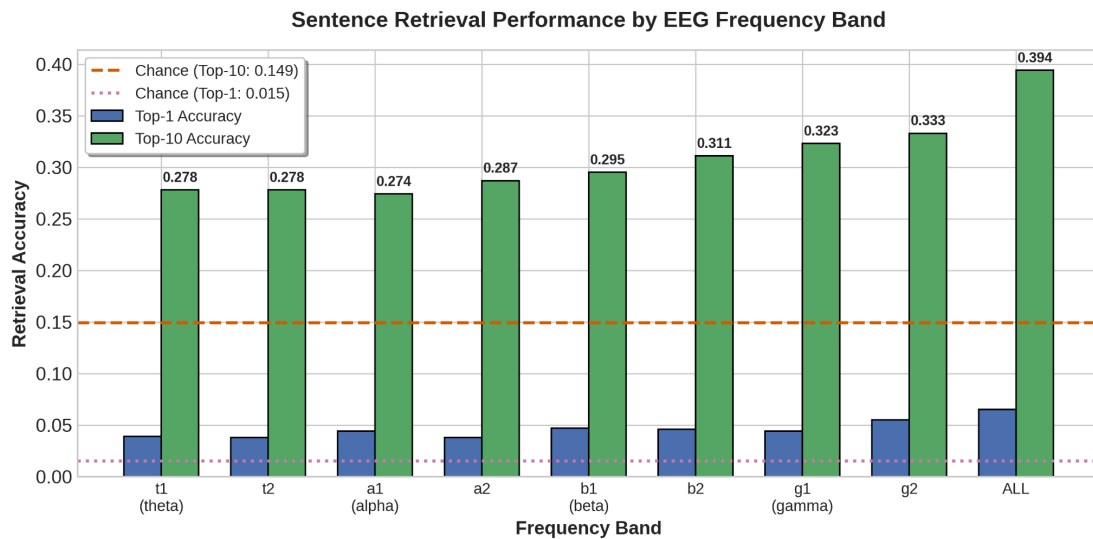
- **Top 1** - was the correct sentence the single #1 guess? chance = $1/67 \approx 0.015$
- **Median rank** - across all recordings, what's the middle position the correct sentence landed at? chance ≈ 34 , the midpoint of 67 (any lower result from 34 is an improvement).

The 4 Sub Experiments:

- **Experiment A - Frequency Bands:**

Tested each of the frequency bands separately (theta, alpha, Beta, gamma)

And gave it a score.



Takeaways:

Every single band beats chance on its own, but combining all 8 together gives the best result. The signal also gets stronger as you go up in frequency — gamma bands carry the most information.

- **Experiment B - Channels:**

tested each of the 105 EEG channels individually. The best single channel barely squeaks above chance (17% vs 14.9% chance).

This means **no single electrode on the scalp carries the signal alone** — it's distributed across the whole head.

- **Experiment C - Word-Level Linguistic Probes:**

The Question being asked on that experiment:

Can the EEG tell you anything about the individual words inside a sentence?

We have 2 probes on that experiment:

1. **The content-vs-function probe:**

A binary classifier. Content words carry meaning — nouns, verbs, adjectives like "Obama," "born," "Hawaii." Function words are grammatical glue — "the," "was," "in," "of." The probe gets one word's EEG and must answer: **content or function?**

The result was accuracy **0.696**, but the majority baseline was **0.699** — meaning if you just blindly guessed "content" every time (because content words are more common), you'd score 0.699 and beat the probe. So the probe learned nothing.

2. **word-length probe:**

It tries to predict how many letters/characters a word has from its EEG. The metric is R^2 , where 0 means "no better than just predicting the average length every time" and 1 means perfect prediction. The result was $R^2 = -0.018$ — negative, which means the probe did slightly worse than the trivial guess-the-mean strategy. Again, no signal.

Takeaways:

The EEG input is **not noise** — genuine, decodable signal exists.

But — and this is the answer that matters for RQ1 — the decodable signal is a **sentence-level fingerprint, not word-level linguistic structure**.

The EEG can tell you which sentence someone read, but carries nothing about the individual words within it — not even surface properties like content-vs-function or word length.

RQ3 - Given runs fail, how and why do they fail?

E3 - when a run scores ~0, how is it failing?

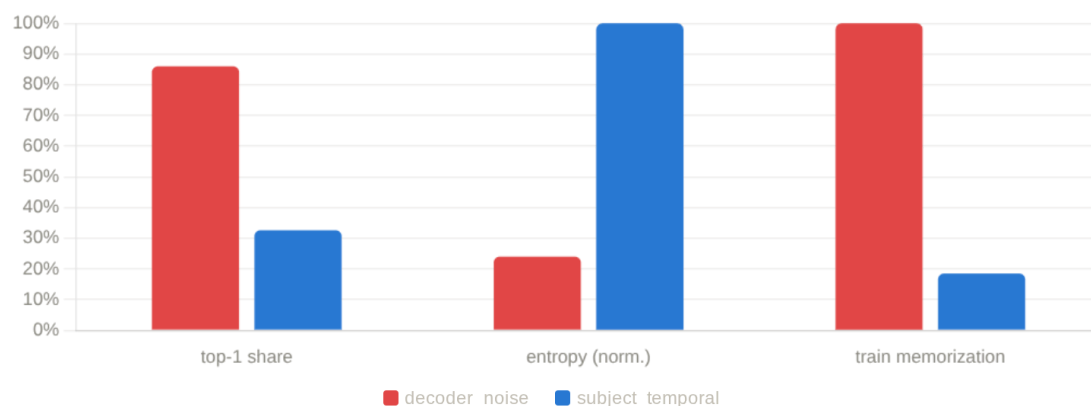
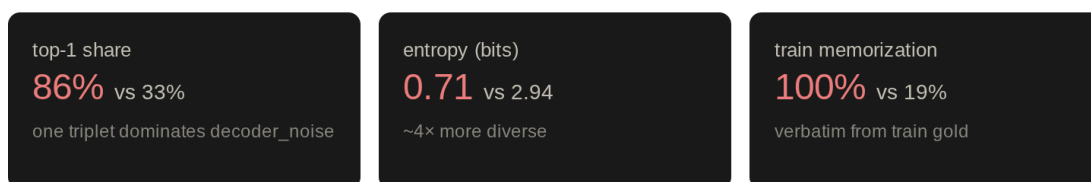
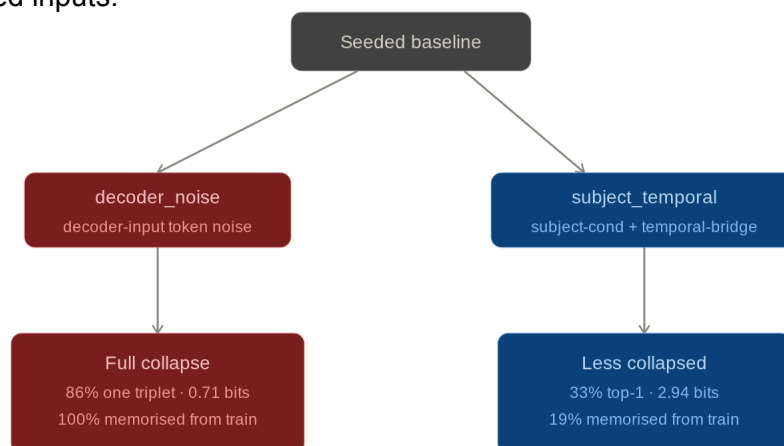
The experiment checks two run log files and decides if the failure was due to collapse-to-memorized-output or varied-but-wrong-output.

What did it test:

E3 is looking at the log files of 2 train runs:

- **Subject_temporal** - a mixture of 2 variants: **subject-cond** & **temporal-bridge**
 - **Subject-cond** - adds a per-subject embedding.
 - **Temporal-bridge** - adds a temporal Transformer pre-encoder in the bridge, so the model processes the EEG's time structure before handing it to the decoder
- **Decoder noise** - run with added noise to the decoder's input tokens during training- in order to regularize and prevent overfitting, and make the decoder rely on the EEG and not just the teacher-forced inputs.

Results:



- **Top-1-share:** how dominant is the single most common output?
what fraction of all 1246 predictions is accounted for by the single most frequent triplet
- **Pred_entropy_bits:** how spread out is the whole output distribution?
Top-1 share only looks at the winner. entropy summarises the entire distribution in one number.
Shannon entropy over the distribution of predicted triplets: 0 bits means the model always emits exactly one thing (perfectly predictable, zero information), and higher bits means the output is spread across more, more-evenly-used options. It's the complement to top-1 share — you could imagine a model with a modest top-1 share but a long thin tail, and entropy catches that where top-1 wouldn't.
- **Train_memorization:** is the output decoded, or copied?
train_memorization is the fraction of predicted triplet-tuples that appear **verbatim in the TRAIN gold set**.
A model that is genuinely (if badly) decoding EEG should produce plenty of triplets that were never in training — it's constructing them from the signal. A model that has given up on the EEG and is just parroting labels it saw during training will echo train tuples at a very high rate.

As we can see in the Graph, the subject temporal is much less collapsed than the decoder_noise version, according to those given metrics.

E2 - Is the decoder's output actually a function of the EEG it's given, or would it produce the same thing regardless?

The experiment uses decoder_noise and subject_temporal checkpoints and asks:
Is the decoder's output actually a function of the EEG it's given, or would it produce the same thing regardless?

How:

For every one of the 1246 test sentences, the same checkpoint was run three times, changing only the EEG fed to the bridge:

- **Real** - the sentence's own true EEG. This is the baseline
- **Swap** - the EEG from a different sentence, assigned via a derangement (a shuffle guaranteed to have no sentence paired with its own EEG). So, sample #5 receives sample #17's actual brain signal, not its own.
- **Zero** - the EEG tensor replaced with all zeros. Importantly, the attention mask was left unchanged - the model still looks at those positions, it just finds nothing there. This isolates content ablation from attention-pattern ablation.

What Is Measured:

- **F1** - which was 0 everywhere - no good.
- **Identical_to_real** - the fraction of samples where the *exact token sequence* generated matches what the same checkpoint produced under the real-EEG condition.

Results:

	swap	zero
decoder_noise output changed	56.5%	53.9%
subject_temporal output changed	92.8%	100%

Conclusion - The decoder is genuinely EEG-sensitive, not EEG-blind:

If the model had learned to ignore the EEG entirely (just memorize and regurgitate), swapping or zeroing it should have left the output essentially unchanged.

RQ4 - Is the frozen BART ceiling a *design/data* problem or a *fundamental* problem?

Why do we ask this question:

We freeze BART. Now the bridge is the only thing that learns. We train it. Test F1 $\approx$ 0. So we have a null result — and a null result is ambiguous. Two very different stories explain it:

- **Story A (design problem):** the bridge is just *badly built*. It's a single Linear+LayerNorm+Dropout. Of course it fails. It has no notion of who's reading (subject variance swamps the signal), no sequence modelling (each word is projected in isolation), and it's supervised by a pooled objective that collapses the sentence to one vector. Fix any of those and F1 climbs. **The ceiling is an artifact of your architecture.**
- **Story B (fundamental problem):** the ceiling is a property of the *regime* — EEG at ZuCo fidelity, pushed through a frozen REBEL, at this data scale. The bridge's design is irrelevant because no bridge, however well-built, can get enough through. **The ceiling is real.**

E5 - Can architecture lift the frozen ceiling?

This experiment includes running 8 separate from-scratch training runs.(2x2x2 grid)
Each cell is a full 60-epoch training job with its own checkpoint.

Fixed attributes:

--freeze_bart, --bridge_layers 2, --align_weight 1.0, --align_temp 0.07, --seed 42, 60 epochs,
early stop after 25 epochs - identical across all 8

The 3 Variations:

- **subject embedding:**
Additive to the embeddings(Bridge output), each subject adds a different embedding
- **temporal bridge:**
Insert a 2-layer Transformer (8 heads) into the bridge, so word positions attend to each other *before* being handed to BART
- **alignment mode (pooled vs perword):**
There's an auxiliary contrastive loss (InfoNCE) that pushes the bridge's output to resemble what REBEL's *text* encoder would produce for the same sentence. A teacher signal.
 - **pooled** — collapse the whole sentence to one vector on both sides, then InfoNCE.
Coarse: it only asks "*does this EEG's overall gist match this sentence's overall gist?*"
 - **perword** — no collapsing. InfoNCE at every position: word 1's EEG vector should match word 1's text vector, word 2's to word 2's, etc. Dense, position-wise supervision.

Results:

#	subj	temp	align	test F1	best val F1 (ep)	align final
1	off	off	pooled	0.0040	0.0015 (39)	2.8732
2	on	off	pooled	0.0083	0.0014 (31)	2.8803
3	off	on	pooled	0.0029	0.0000 (1)	2.8478
4	on	on	pooled	0.0000	0.0000 (1)	2.9092
5	off	off	perword	0.0000	0.0000 (1)	6.2145
6	on	off	perword	0.0000	0.0000 (1)	6.2140
7	off	on	perword	0.0000	0.0000 (1)	6.1751
8	on	on	perword	0.0000	0.0000 (1)	6.1555

Conclusions:

Architecture doesn't move F1, and we don't know why

E6 - Does more data lift the F1 needle?

E5 closed the architecture question: under a frozen REBEL decoder, no architectural axis we tried — subject embedding, temporal bridge, per-word alignment, or any combination — lifted test F1 off the floor. That left exactly one uncontrolled lever that could still explain the failure benignly. If the bridge simply hasn't seen enough sentences, then the ~0 ceiling is a sample-size artifact, and the thesis story is "data-limited, not fundamental" — the fix would be more subjects or a ZuCo-1 merge. E6 asks:

Does test F1 climb as the model sees more training sentences, or is the curve flat at the floor?

Decision rule: A curve still rising at `frac = 1.0` reframes the ceiling as data-limited. A flat curve means architecture (E5) and data (E6) are both exhausted, and by elimination, the binding constraint is EEG signal fidelity itself at word granularity — the E1 cliff.

What we manipulated:

We held the best E5 cell completely fixed — `--n_subject_buckets 64` on, temporal bridge off, `--align_mode pooled`, `--freeze_bart`, seed 42, 60 epochs, patience 25 — and varied only `--train_sentence_frac` $\in \{0.10, 0.25, 0.50, 1.00\}$.

Three design choices make the curve mean what it claims. First, subsampling is **by unique sentence, not by sample** (`_select_sentence_fraction`, deterministic by seed): the same sentence read by multiple subjects stays together, so the split invariant holds and the curve measures the effect of seeing fewer *sentences* — i.e. less linguistic diversity — rather than fewer copies of the same one. Second, **val/test are always left full**, so all four points are scored on the identical 1246-sample test set and the F1 numbers are directly comparable. Third, `frac 1.00` is E5 cell 2 re-run under byte-identical code and seed, which anchors the top of the curve and confirms the E6 harness isn't silently doing something different from E5.

What we got:

Test F1 across the curve: 0.0006 (frac 0.10, ~1015 sentences), 0.0000 (0.25), 0.0023 (0.50), 0.0083 (1.00, 10216 sentences). A 10× increase in training data buys an absolute gain of ~0.008 — which is *the same magnitude as the entire E5 architecture grid spread*. The whole curve lives inside one noise band. The aggregator flags a +0.0060 slope over the top half (0.50 → 1.00), which in isolation could read as "still climbing," but that reading dies on the fact that the curve is **non-monotone**: frac 0.25 scores 0.0000, *below* frac 0.10. A genuine data-scaling curve does not go down when you double the data. Corroborating this, the only two points with best-val past epoch 1 (frac 0.10 at ep 16, frac 1.00 at ep 31) are also the only two with any nonzero F1 — a handful of lucky tuple matches over 1246 test samples. The right phrase is "marginally less degenerate," not "working."

What this tells us:

E5 killed "bad architecture." E6 killed "not enough data."

That leaves one suspect: the EEG signal at word granularity doesn't carry enough information to decode. And that suspect was already identified independently and directly by E1 (fidelity cliff) and E3 (prediction collapse). Four experiments, two of them direct and two by elimination, all converge on the same bottleneck — which is why this negative result holds up rather than reading as "we couldn't get it to work."